

28-02-2026

Agenda:

pandas

Git

Group-1 Group-2 Group-3 Group-4

N1 N4 N5 N6 → Dataset
N2 ↓ ↓ ↓
N3 attributes attributes attributes

each of the students has their own attributes, that is they belong together.

(All total students)

6 → attributes → group

If I pass each person's attributes
person → attributes

ML-model
(learn based on 6 student)

My ML model should tell me that they belongs to which group

1
2
3
4

this is called as

Classification

ML doesn't understand text, so it converted to numbers

(1, 1, 1)

Group-1
Group-2
Group-3
Group-4

ML →

1
2
3
4

encoding

Group → categorical (numerical)

"A"

"B" → character / string → pandas → D → A

"C"

"D"

A . . . 2

D
C
B
A

16

strings doesn't hold the sequence but number does.

"Two" > "first" : True

String $\rightarrow$ converted $\rightarrow$ dtype $\rightarrow$ to handle order
 $\downarrow$
 logic

event	date	ticket_id	price
concert	2028-02-02	—	—
—	—	—	—
—	—	—	—
conference	—	—	—

Data analysis $\rightarrow$ conclusion $\rightarrow$

rows → date → 2026-02-02

```

graph TD
    string --> year
    string --> month
    string --> day
    string --> weekday

```

row) $\rightarrow$ 02-02-2026 $\leftrightarrow$ 28-02-2026

→
str

Jan-1 → new year

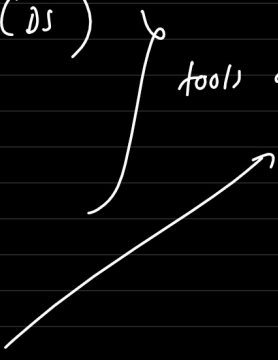
→ today Jan-30

New year — today → -

holi — today → +

python + OOP (DS)
numpy (DS)
pandas (DS)
git (development)
Data visualisation

tools of data science

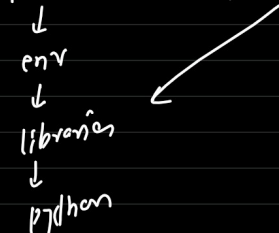


DS :- understanding data →

→ data —
→ analysis →
→ fix →
→ getting started →
→ plotting →

Git :- Tool to manage code

UV → project + dependencies
↓
env
↓
libraries
↓
python

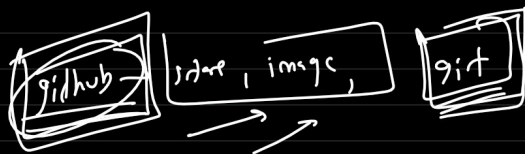
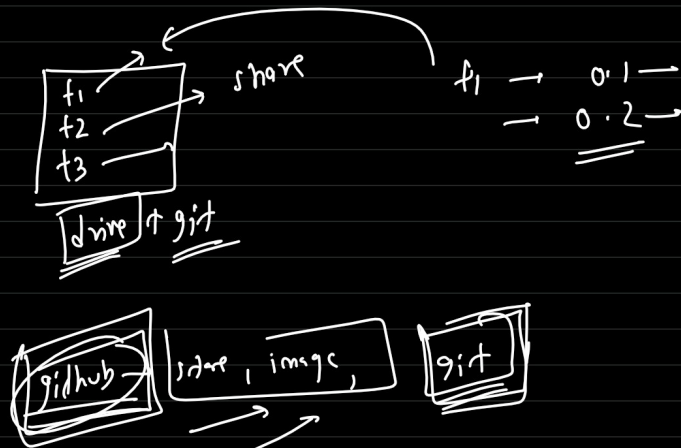


crit → project → p1 ← code
 → p2 ← code
 → p3 ← code
 → p4 ← code

p1 → Today → + 10000 lines → working → save → commit cp
 → Day 11 → + 10 lines → error
 update 10000 lines → error → save → commit cp
 → commit
git
 git → diff → file

Version : 0.1 → code
 → : 0.2 → git ^ Version control tool
 — : 0.3

git → Tool
 github → central repository of your code + git

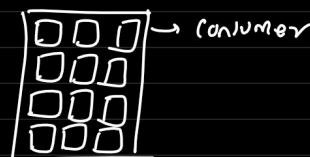


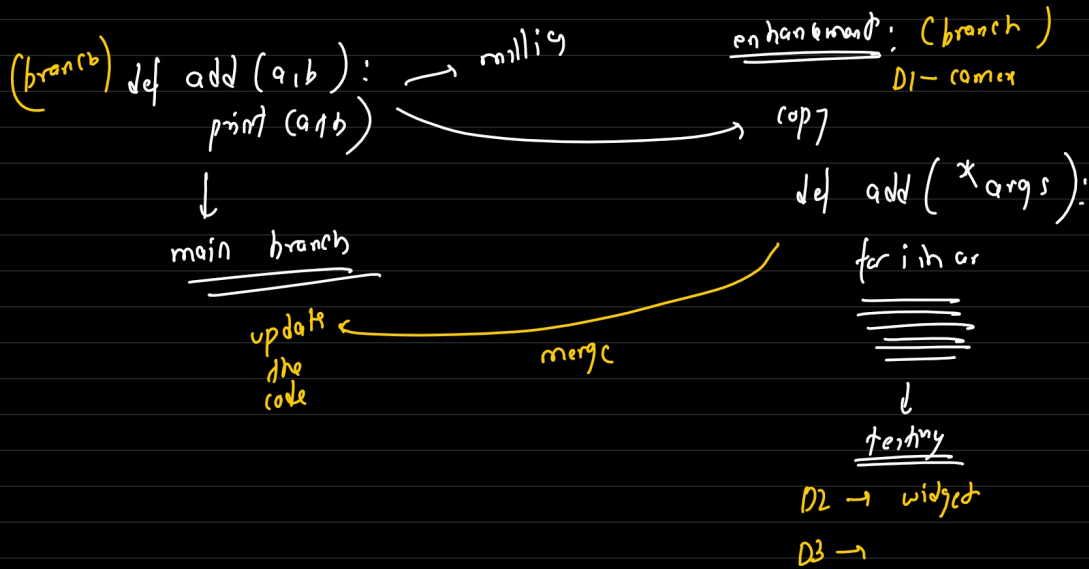
git + online

project → ios 16 →

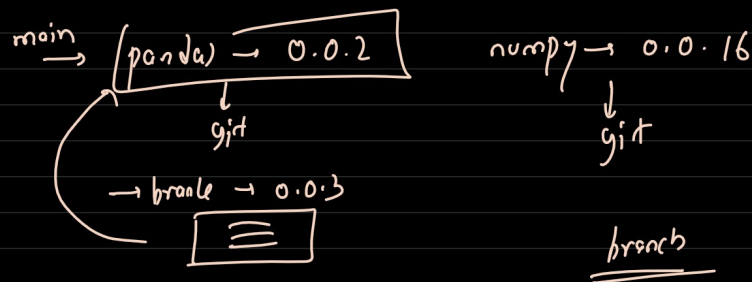
million of people
 ios 16 → project → copy

↓
 work





- manage code by creating checkpoints, version
- separate code by branches



online: repo → github

D1:

local → git
code

↑ git init

→ git add .

→ git commit -m "abc"

→ git branch -M main

1st time thing

→ git remote add origin
url of the repo

→ git push -u origin main

D2:

git clone url-of-repo

D2: → uploaded files using github UI → .txt → github → online

github online

f1

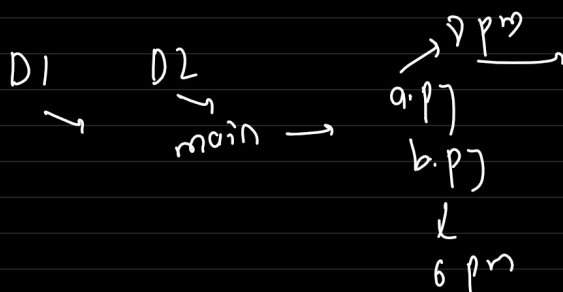
f2

local repo

f1

→ git pull origin branch-name

↓
main



log of

↓

log in

git pull

project-1

study-1

pandas
numpy

.venv

.py

pandas

.venv

.py